

SCHOOL OF ENGINEERING,
COCHIN UNIVERSITY OF SCIENCE AND TECHNOLOGY
B. TECH DEGREE II SEMESTER FIRST INTERNAL EXAMINATIONS, APRIL 2024
23-200-0201A: COMPUTER PROGRAMMING AND PROBLEM SOLVING (CE-A)
(2023 Scheme)

TIME: 2 hrs

MAX. MARKS: 30

Course Outcomes

On completion of this course the student will be able to

CO1	Identify main components of computer system and explain its working
CO2	Develop flow chart and algorithms for computational problems
CO3	Write the syntax of various constructs of C language

PART A

(Answer all questions)

(5 x 2= 10 Marks)

Q.No.	Questions	Marks	BL	CO	PO
I (a)	Write a short note on memory of a computer system	2	L1	1	1
(b)	Compare and contrast compilers and interpreters	2	L2	1	1
(c)	Draw a flow chart to find the average of three numbers	2	L2	2	1
(d)	Distinguish between different categories of programming languages	2	L4	1	2
(e)	Predict the out of the following code <pre>#include<stdio.h> int main() { int a=10,b=5; printf("%d\n", a++ + ++b); printf("%d\n", a); printf("%d\n", b- - - a); printf("%d\n", a); printf("%d\n", b); return 0; } Justify your answer.</pre>	2	L5	3	3

PART B

(Answer the following questions
choosing either the first OR second option)

(1 x 10= 10 Marks)					
Q.No.	Questions	Marks	BL	CO	PO
II (a)	Draw the block diagram of the digital computer and Explain each unit in detail.	6	L5	1	3
(b)	Explain about Application software and System software	4	L2	1	2
OR					
III (a)	Write a C program to find factorial of a number	5	L1	2	2
(b)	Write a C program to find fibonacci series	5	L1	1	2
(1 x 10= 10 Marks)					
IV (a)	Write a C program to find prime numbers of given range	5	L4	3	3
(b)	Explain the following tokens with examples: keywords, identifiers and constants.	5	L3	3	3
OR					
V (a)	Write a C program to reverse a number	5	L3	3	4
(b)	Write a C program to find sum of digits of a number	5	L6	3	4

Bloom's Taxonomy Levels

L1 - %	L2 - %	L3 - %	L4 - %	L5 - %	L6 - %
30	20	25	17.5	20	12.5

